Step 2a - H6 - Millers enzyme inactivation without steam

Market scan

Market size

The addressable frame is the global flour milling industry: ~285 Mt of wheat/meslin flour consumed in 2024 (~$150-190B value, growing ~1-2%/yr), ~450 Mt total flour including other grains. Enzyme inactivation (stabilization) is relevant mainly to whole-grain flour, bran, germ, oats and rice bran - a niche of the ~$12B/yr milling-equipment market, but growing with whole-grain and clean-label demand (~6% CAGR in wheat germ flour). The equipment niche itself is likely a few hundred M$/yr globally, order of magnitude.

Region	Size ($B)	Region	Size ($B)
Europe	~EUR 15-20B flour milling turnover (EU-27); ~35 Mt flour/yr; ~25% of world flour consumption	Asia	largest region, ~45% of world flour consumption; China ~60-65 Mt flour (2,000+ mills), India very large but mostly small chakki mills; ~$60-80B flour value
America	~12% of world flour consumption; US ~20-21 Mt flour, highly consolidated; ~$25-35B flour value incl. LatAm	Africa	~15% of world flour consumption, fastest growth; mix of large industrial mills (Egypt, Nigeria) and small mills; ~$20-30B flour value

Target customers

Very fragmented globally but easy to segment. Europe has >3,000 mills (mostly SMEs) reachable via the European Flour Millers association and national bodies; the US is extremely consolidated (~160 wheat mills, top 21 companies = 96% capacity) so a handful of groups (Ardent, ADM, Mennel...) unlock the market; China has 2,000+ mills incl. ~300 large modern plants; India has ~29,000 mostly small chakki mills, hard to reach. Realistic buyers are millers making whole-grain flour, stabilized bran/germ, and oat/rice millers - roughly 10-20% of mills care about stabilization today.

Customer size	Approx. number	Market share (in %)
Large	~500-1,000 large industrial mills worldwide (e.g. China ~300 large plants; US top ~160 mills; European groups)	~60-70% of world flour volume; near-total share in US (96% via top 21 cos.)
Medium	~2,000-4,000 mid-size mills (bulk of EU-27's >3,000, Turkey ~680, Mexico ~1,100)	~20-30% of volume; strongest in Europe
Small	~30,000+ small/chakki mills, concentrated in India (~29k) and Africa	~10-15% of volume; low willingness/ability to pay

Players offering a solution today

Supplier side is split between a few dominant milling OEMs and niche thermal specialists. Bühler dominates hydrothermal stabilization (Granotherm kiln) with Satake, Alapala, Ocrim, Streckel & Schrader offering similar steam/kiln systems. Rice-bran stabilization in Asia is dominated by dry-extrusion suppliers

(Insta-Pro, Clextral, local Indian/Chinese OEMs). Dedicated industrial-microwave suppliers (SAIREM, Pueschner, Chinese tunnel makers) exist but offer almost no milling-specific enzyme-inactivation product - the steam-free electric niche is essentially open.

Player	Type of solution	Market share (in %)
Large (Buehler, Satake, Alapala, Ocrim)	Steam/hydrothermal kilns and stabilization lines integrated in new mills (Granotherm DNCB, oat kilners)	~60-70% of installed stabilization capacity, est.
Medium (Revtech, Streckel & Schrader, Clextral/Insta-Pro/Wenger extrusion, Vomm)	Specialist continuous stabilizers (vibratory towers, kilns) and dry extruders/expanders for bran/oilseed stabilization	~25-30%, est.
Small (SAIREM, Pueschner, AMTek, Chinese microwave OEMs)	Generic industrial microwave/dielectric heating equipment, mostly dryers and tempering - little enzyme-inactivation-specific product for mills	<10%, est.

Regulation

Regulation	Jurisdiction	What it means for us
ATEX Directive 2014/34/EU + ATEX workplace Directive 1999/92/EC (UK: DSEAR)	EU/UK	Flour/bran dust creates Zone 20-22 explosive atmospheres inside mills; any electric/microwave machine installed in a milling line must be ATEX-certified or placed outside zoned areas - adds design cost and is a real barrier for a small new OEM.
Regulation (EC) 852/2004 / FDA FSMA preventive controls	EU / US	Thermal treatment used as a process control must be validated and monitored; a microwave line needs documented uniform enzyme-inactivation (peroxidase often used as marker) across the product bed.
Machinery Directive 2006/42/EC + EN 55011 / ISM-band limits	EU	Microwave generators run at 2,450 MHz ISM band; leakage and EMC limits apply; standard CE-marking burden.
Novel Food Regulation (EU) 2015/2283	EU	Microwave-treated bran/germ is not considered novel (heating is a conventional process), so no market-authorization hurdle for the treated ingredient.

Energy/decarbonization policy (EU ETS, national electrification incentives)	EU	Electrification of industrial heat is subsidized/favored vs. gas-fired steam boilers - tailwind for an all-electric machine, though EU industrial electricity prices (~0.10-0.15 EUR/kWh vs gas ~0.03-0.05 EUR/kWh) hurt OPEX parity.
Food-contact materials Reg (EC) 1935/2004	EU	Product-contact surfaces must be compliant stainless/food-grade - same burden as incumbents, no extra hurdle.

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis